Unit: 6 Unit 5: Linear Equations
Topic: Perpendicular Lines

Name: ____________
Date: __________

Advanced (Carolina Reaper)

Q1. What is the slope of a line perpendicular to $y = 2x + 1$?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 0

Q2. If two lines have slopes m_1 and m_2, what is the condition for the lines to be perpendicular?
(A) $m_1 = m_2$
(B) $m_1 = -m_2$
(C) $m_1 \cdot m_2 = -1$
(D) $m_1 + m_2 = 0$
(E) $m_1 - m_2 = 1$

Q3. A line passes through $(1, 2)$ and is perpendicular to $y = 3x - 1$. What is the equation of this line?
(A) $y = -\frac{1}{3}x + \frac{7}{3}$
(B) $y = \frac{1}{3}x + \frac{5}{3}$
(C) $y = 3x - 1$
(D) $y = -3x + 5$
(E) $y = \frac{1}{3}x - \frac{1}{3}$

Q4. In a city, the main street has a slope of -2. What is the slope of a street perpendicular to it?
(A) $\frac{1}{2}$
(B) $-\frac{1}{2}$
(C) 2
(D) -2
(E) $\frac{1}{4}$

Q5. A hiker is walking on a trail that has a slope of $\frac{3}{4}$. What is the slope of a trail perpendicular to it?
(A) $-\frac{4}{3}$
(B) $\frac{4}{3}$
(C) $\frac{3}{4}$
(D) $-\frac{3}{4}$
(E) 1

Q6. A bookshelf has a support beam with a slope of -1. What is the slope of a beam perpendicular to it?
(A) 1
(B) -1
(C) 0
(D) $\frac{1}{2}$
(E) 2

Q7. A clock has hour and minute hands that form an angle. If the minute hand has a slope of 2, what is the slope of the hour hand if it is perpendicular to the minute hand?
(A) $-\frac{1}{2}$
(B) $\frac{1}{2}$
(C) 2
(D) -2
(E) 0

Q8. A bike trail has a slope of $\frac{2}{5}$. What is the slope of a trail perpendicular to it?
(A) $-\frac{5}{2}$
(B) $\frac{5}{2}$
(C) $\frac{2}{5}$
(D) $-\frac{2}{5}$
(E) 1

Open Ended Questions (FRQ)

Q9. A skateboard ramp has a slope of -3. If a skateboarder wants to design a perpendicular ramp, what is the equation of the new ramp if it passes through $(2, 1)$? Show your work and provide the equation in slope-intercept form $y = mx + b$.

Q10. A city planner wants to design a new road that is perpendicular to an existing road with a slope of $\frac{1}{4}$. If the new road passes through $(1, 3)$, what is the equation of the new road? Show your work and provide the equation in slope-intercept form $y = mx + b$.

Chef's Tips

- Provide clear instructions for each question
- Ensure students have the necessary materials and equipment
- Encourage students to show their work and provide explanations for their answers